

Paper 2, Paper 4 Exam Paper

Candidate Name: _____

Total Marks: 48

Question 1

(a) Simplify each of the following expressions, leaving your answers in index form.

(i) $3^4 \times 3^2$

..... [1]

(ii) $\frac{5^7}{5^4}$

..... [1]

(iii) $(2^3)^4$

..... [1]

Question 2

(a) Evaluate each of the following numerical expressions. Give your answers as integers or fractions in their simplest form.

(i) 7^0

..... [1]

(ii) 4^{-2}

..... [1]

(iii) 3×2^{-3}

..... [1]

Question 3

(a) Find the exact value of each expression.

(i) $25^{\frac{1}{2}}$

..... [1]

(ii) $27^{\frac{1}{3}}$

..... [1]

(iii) $16^{\frac{3}{4}}$

..... [2]

Question 4

(a) Simplify the following operations completely.

(i) $2^{-4} \times 2^6$

..... [1]

(ii) $\frac{6^2}{6^{-1}}$

..... [1]

(iii) Show that the expression $\left(\frac{2}{3}\right)^{-2}$ reduces exactly to $2\frac{1}{4}$.

[2]

Question 5

(a) Work out the value of each expression, giving your answer as a single integer or a simplified fraction.

(i) $8^{-\frac{1}{3}}$

..... [2]

(ii) $\left(\frac{9}{4}\right)^{-\frac{1}{2}}$

..... [2]

Question 6

- (a) Evaluate each of the following expressions completely, giving your final answer as an exact integer or a fraction in its simplest form.

(i) $\left(\frac{1}{8}\right)^{-\frac{2}{3}}$

..... [2]

(ii) $125^{\frac{1}{3}} \times 2^{-3}$

..... [2]

Question 7

- (a) Solve the following index equations for x .

(i) $3^x = \frac{1}{81}$

..... [2]

(ii) $4^{2x-1} = 32$

..... [2]

Question 8

(a) Simplify the operational expressions.

(i) Find the exact value of the expression.

$$\frac{2^5 \times 6^3}{3^2 \times 4^3}$$

..... [3]

(ii) Show that the expression reduces exactly to an integer.

$$\frac{\left(5\frac{1}{2}\right)^4 \times 10^2}{2^2 \times 5^3}$$

[2]

Question 9

(a) Work out the value of the following composite expressions.

(i) $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$

..... [2]

(ii) $\left(4^{\frac{3}{2}} + 27^{\frac{2}{3}}\right)^{-1}$

..... [3]

Question 10

- (a) Evaluate each of the operations without using a calculator, leaving answers as standard rational values.

(i) $\left(2\frac{7}{9}\right)^{-\frac{1}{2}}$

..... [2]

(ii) $\frac{3^{-5} \times 9^2}{27^{-1}}$

..... [2]

Question 11

Solve the simultaneous index equations for x and y .

$$2^x \times 4^y = \frac{1}{8}$$

$$27^x \times 3^y = 9^{\frac{1}{2}}$$

..... [4]

Question 12

- (a) An advanced engineering formula determines the operational efficiency metric, E , based on an input index variable x such that:

$$E = \frac{243^{\frac{x}{5}} \times 4^{-x}}{3^{x-1}}$$

- (i) Show analytically that the expression for E simplifies exactly to 3×4^{-x} .

[2]

- (ii) Find the value of E when $x = 2.7$, correct to 3 significant figures.

..... [2]

Topic Checklist & Self-Reflection

Review the topics below and circle your confidence level.

Question	Topic Details	Confidence
Question 1	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 2	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 3	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 4	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 5	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 6	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 7	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 8	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 9	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 10	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 11	Arithmetic Indices and Power Rules	[Low] [Med] [High]
Question 12	Arithmetic Indices and Power Rules	[Low] [Med] [High]